Worked Answer Key: Practice Test 5.1–5.3

1. (5.1) $f(x) = x^2 - 1$, $g(x) = 3x + 2$.

- (a) $f(g(1))$: inside first, $g(1) = 3 + 2 = 5$. Then $f(5) = 25 - 1 = \boxed{24}$.
- (b) $g(f(1))$: $f(1) = 1 - 1 = 0$. Then $g(0) = \boxed{2}$.
- (c) $f(f(2))$: $f(2) = 3$. Then $f(3) = 9 - 1 = \boxed{8}$.
- (d) Replace x in f with $(3x+2)$: $f(g(x)) = (3x+2)^2 - 1 = 9x^2 + 12x + 4 - 1 = \boxed{9x^2 + 12x + 3}$.
- (e) Replace x in g with $(x^2 - 1)$: $g(f(x)) = 3(x^2 - 1) + 2 = 3x^2 - 3 + 2 = \boxed{3x^2 - 1}$.
- Check (d) with (a): $9 + 12 + 3 = 24$ ✓.

2. (5.1) Table lookups. Find the inside value in the g (or f) row, then look up **that number** in the x row.

- $f(g(1))$: $g(1) = 2$, then $f(2) = \boxed{4}$.
- $g(f(1))$: $f(1) = 3$, then $g(3) = \boxed{4}$.
- $f(f(3))$: $f(3) = 1$, then $f(1) = \boxed{3}$.
- $g(g(4))$: $g(4) = 3$, then $g(3) = \boxed{4}$.

3. (5.1) Composites and domains.

- (a) $f(g(x)) = \sqrt{x - 5}$. The square root needs $x - 5 \geq 0$, so the domain is $\boxed{x \geq 5}$. $g(f(x)) = \sqrt{x} - 5$. Only $\sqrt{x}$ needs care: $\boxed{x \geq 0}$.
- (b) $f(g(x)) = \dfrac{2}{\frac{4}{x} - 1}$. Multiply the top and bottom by x: $\dfrac{2x}{4 - x}$. Domain, checked **before** simplifying:
  1. $g(x) = \frac{4}{x}$ needs $x \neq 0$.
  2. f can't take an input of 1: $\frac{4}{x} \neq 1$, so $x \neq 4$.

  Domain: $\boxed{\text{all reals except } 0 \text{ and } 4}$. The simplified $\frac{2x}{4-x}$ looks fine at $x = 0$, but the original isn't. That's why you find the domain first.

4. (5.1) Decompose: find the inside piece and the outside action.

- (a) Inside: $g(x) = 2x + 5$. Outside: "cube it," $f(x) = x^3$.
- (b) Inside: $g(x) = x^2 + 1$. Outside: "square root it," $f(x) = \sqrt{x}$.
- (c) Inside: $g(x) = x - 7$. Outside: "1 over it," $f(x) = \frac{1}{x}$.
- Check: plug g into f and you get h back. Other splits can also be correct.

5. (5.1) $n(t) = 25t$ shirts in t hours; $C(n) = 10n + 50$ dollars.

- (a) Plug $n(t)$ in for n: $C(n(t)) = 10(25t) + 50 = \boxed{250t + 50}$. Units: hours → shirts → dollars, so this is **the cost in dollars of t hours of production**.
- (b) $n(4) = 100$ shirts, so $C(100) = 1000 + 50 = \boxed{1050}$. Four hours of production (100 shirts) costs \$1,050. Check with (a): $250(4) + 50 = 1050$ ✓.

6. (5.2) One-to-one?

- (a) The output 4 comes from both $x = 1$ and $x = 3$. **Not one-to-one.**
- (b) Outputs 2, 3, 7 are all different. **One-to-one.**
- (c) $y = x^2$: $2^2 = (-2)^2 = 4$, so it fails the horizontal line test. **No.**
- (d) $y = x^3$: always increasing, so every horizontal line hits once. **Yes.**
- (e) $y = |x|$: $|3| = |-3|$. **No.**
- (f) $y = 2^x$: always increasing. **Yes.**

7. (5.2) f through $(-2, 1)$, $(0, 3)$, $(4, 7)$; domain $[-2, 4]$, range $[1, 7]$.

- (a) Swap each pair: $\boxed{(1, -2),\ (3, 0),\ (7, 4)}$.
- (b) $f(5) = 11$ means $f^{-1}(11) = \boxed{5}$ (input and output trade places).
- (c) Domain and range swap: f^{-1} has domain $\boxed{[1, 7]}$ and range $\boxed{[-2, 4]}$.

8. (5.2) Read the table backwards: find the output in the $f(x)$ row and read up to x.

- $f^{-1}(11)$: $f(2) = 11$, so $\boxed{2}$.
- $f^{-1}(5)$: $f(0) = 5$, so $\boxed{0}$.
- $f^{-1}(f(3))$: an inverse undoes the function, so this is just $\boxed{3}$. (Or: $f(3) = 14$ and $f^{-1}(14) = 3$.)

9. (5.2) Find the inverse: write $y =$, swap x and y, solve for y.

- (a) $x = 5y - 3$. Add 3: $x + 3 = 5y$. Divide: $\boxed{f^{-1}(x) = \dfrac{x+3}{5}}$.
- (b) $x = y^3 + 4$. Subtract 4: $y^3 = x - 4$. Cube root: $\boxed{f^{-1}(x) = \sqrt[3]{x-4}}$.
- (c) $x = \dfrac{y+1}{y-2}$.
  1. Multiply both sides by $(y-2)$: $x(y-2) = y + 1$.
  2. Distribute: $xy - 2x = y + 1$.
  3. Get the y-terms on one side: $xy - y = 2x + 1$.
  4. Factor out y: $y(x-1) = 2x + 1$.
  5. Divide: $\boxed{f^{-1}(x) = \dfrac{2x+1}{x-1}}$.

  Domain of f: $x \neq 2$. Range of f = domain of f^{-1}: $y \neq 1$. (It matches: the horizontal asymptote of f is $y = 1$.)

10. (5.2) $f(x) = 2x - 6$, $g(x) = \frac{x}{2} + 3$.

- $f(g(x)) = 2\left(\frac{x}{2} + 3\right) - 6 = x + 6 - 6 = x$ ✓
- $g(f(x)) = \dfrac{2x-6}{2} + 3 = x - 3 + 3 = x$ ✓
- Both give x, so they're inverses. (You need to check both directions.)

11. (5.2) $f(x) = (x-2)^2$.

- (a) The vertex is at $x = 2$, and the right half is always increasing: restrict to $\boxed{x \geq 2}$.
- (b) $y = (x-2)^2$. Swap: $x = (y-2)^2$. Square root: $\sqrt{x} = y - 2$ (take the **positive** root, because $y \geq 2$ on the restricted piece). Add 2: $\boxed{f^{-1}(x) = 2 + \sqrt{x}}$. The restricted f has domain $x \geq 2$ and range $y \geq 0$, so f^{-1} has **domain** $x \geq 0$ and **range** $y \geq 2$.

12. (5.2) f through $(0,1)$, $(1,2)$, $(2,4)$.

- Swap coordinates: f^{-1} goes through $(1,0)$, $(2,1)$, $(4,2)$.
- On the graph, each point flips across the dashed line $y = x$, landing the same distance from the line on the other side. Points on the line itself (where $x = y$) wouldn't move.
- The sketch should look like two mirror images across $y = x$.

13. (5.3) $f(x) = 3 \cdot 2^x$.

- $f(0) = 3 \cdot 1 = \boxed{3}$ (the initial value a).
- $f(3) = 3 \cdot 8 = \boxed{24}$.
- $f(-1) = 3 \cdot \frac{1}{2} = \boxed{1.5}$ (a negative exponent means 1 over).

14. (5.3)

- (a) $500(1.03)^t$: initial value **500**, $b = 1.03 > 1$ so **growth**, $1.03 - 1 = 0.03$, **3%**.
- (b) $80(0.6)^t$: initial value **80**, $b = 0.6 < 1$ so **decay**, $1 - 0.6 = 0.4$, **40%**.

15. (5.3) $f(x) = 2^{x-1} + 3$.

- $x - 1$ inside: shift **right 1**. $+3$ outside: shift **up 3**.
- Domain: **all reals** (exponentials accept any input).
- The asymptote of 2^x is $y = 0$. Shifting up 3 moves it to $\boxed{y = 3}$. The range moves with it: $\boxed{y > 3}$.
- y-intercept: $f(0) = 2^{-1} + 3 = 0.5 + 3 = \boxed{3.5}$.

16. (5.3) $y = ab^x$ through two points.

- (a) $(0,4)$ gives $a = 4$ directly. $(2,36)$: $4b^2 = 36$, so $b^2 = 9$ and $b = 3$. $\boxed{y = 4 \cdot 3^x}$
- (b) $ab = 10$ and $ab^3 = 2.5$. Divide the second by the first: $b^2 = \frac{2.5}{10} = 0.25$, so $b = 0.5$. Then $a(0.5) = 10$, so $a = 20$. $\boxed{y = 20(0.5)^x}$
- Check (b): $20(0.5)^3 = 20 \cdot \frac{1}{8} = 2.5$ ✓.

17. (5.3) Make the bases match, then set the exponents equal.

- (a) $16 = 2^4$: $2^{x+1} = 2^4$, so $x + 1 = 4$ and $\boxed{x = 3}$.
- (b) $9 = 3^2$ and $27 = 3^3$: $(3^2)^x = 3^3$, so $3^{2x} = 3^3$, $2x = 3$, and $\boxed{x = \frac{3}{2}}$.
- (c) $\frac{1}{2} = 2^{-1}$ and $8 = 2^3$: $2^{-x} = 2^3$, so $-x = 3$ and $\boxed{x = -3}$.
- (d) $81 = 3^4$: $3^{x^2} = 3^4$, so $x^2 = 4$ and $\boxed{x = \pm 2}$. (Don't forget the negative root.)

18. (5.3) \$5,000 at 6% for 5 years.

- Quarterly: $n = 4$, so $A = 5000\left(1 + \frac{0.06}{4}\right)^{4 \cdot 5} = 5000(1.015)^{20} \approx 5000(1.34686) \approx \boxed{\$6{,}734.28}$.
- Continuous: $A = 5000e^{0.06 \cdot 5} = 5000e^{0.3} \approx 5000(1.34986) \approx \boxed{\$6{,}749.29}$.
- Continuous compounding earns a little more, as expected.

19. (5.3) 200 g, half-life 10 years.

- $A(t) = 200\left(\frac{1}{2}\right)^{t/10}$ (the exponent $t/10$ counts how many half-lives have passed).
- 25 years is 2.5 half-lives: $A(25) = 200\left(\frac{1}{2}\right)^{2.5} \approx 200(0.17678) \approx \boxed{35.4 \text{ g}}$.
- Sanity check: 2 half-lives leave 50 g and 3 leave 25 g. 35.4 is between ✓.

20. (5.1-5.3) $f(x) = 2^x$, $g(x) = x - 3$.

- (a) $h(x) = f(g(x)) = 2^{x-3}$. That's 2^x shifted **right 3**.
- (b) **Yes.** 2^{x-3} is always increasing (a shifted exponential with $b = 2 > 1$), so it passes the horizontal line test.
- (c) $h(5) = 2^{5-3} = 2^2 = 4$, so $(5, 4)$ is on h. Swap the coordinates: $\boxed{(4, 5)}$ is on h^{-1}.
- (d) h has horizontal asymptote $y = 0$ (shifting right doesn't move it). Reflecting over $y = x$ turns a horizontal line $y = 0$ into the vertical line $\boxed{x = 0}$, which h^{-1} approaches. (Preview: $h^{-1}(x) = \log_2 x + 3$.)